

DATABASE PROJECT (Furniture database)

Query 1(Find the total quantity, gross amount, discount, and income)

SELECT

SUM(quantity) AS total_quantity,

SUM(quantity * unit_price) AS gross_sales,

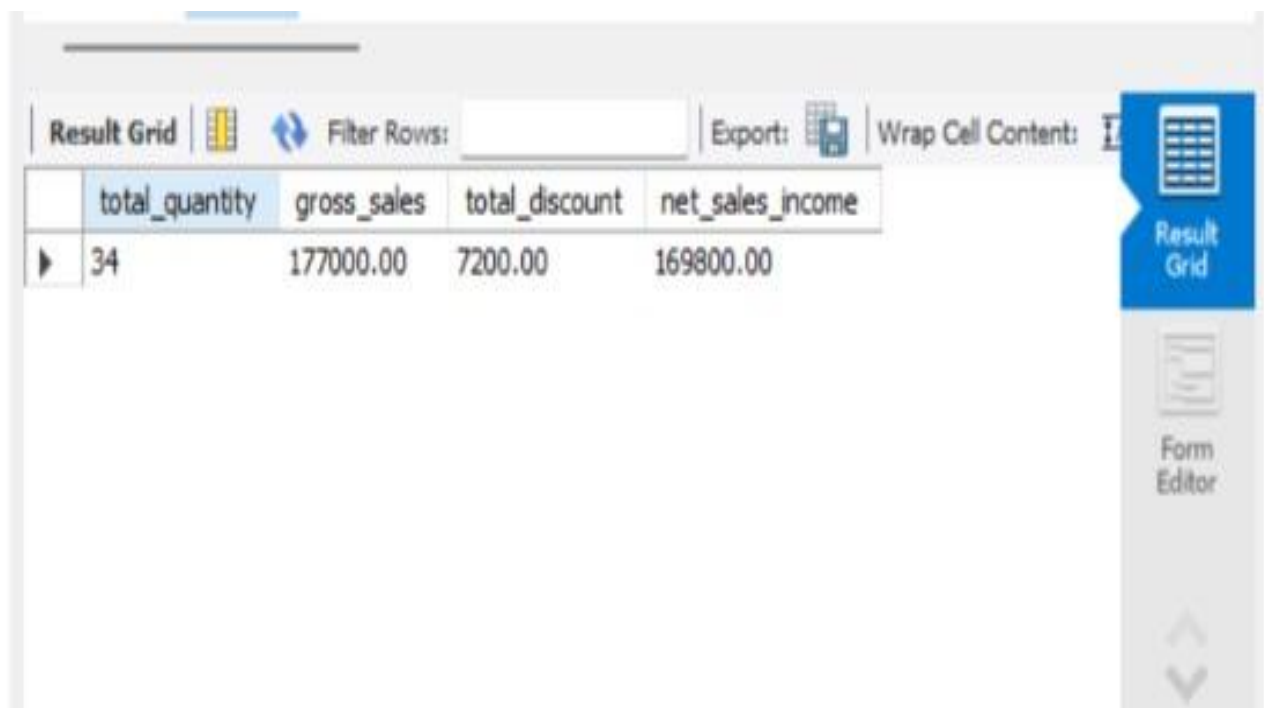
SUM(discount_amount) AS total_discount,

SUM(quantity * unit_price - discount_amount)

AS net_sales_income

FROM Sales;

Result



The screenshot shows a database query result interface. At the top, there is a toolbar with options: 'Result Grid' (selected), 'Filter Rows:', 'Export:', and 'Wrap Cell Content:'. Below the toolbar is a table with four columns: 'total_quantity', 'gross_sales', 'total_discount', and 'net_sales_income'. The first row of data shows the values 34, 177000.00, 7200.00, and 169800.00 respectively. On the right side of the interface, there is a vertical sidebar with a 'Result Grid' button and a 'Form Editor' button.

	total_quantity	gross_sales	total_discount	net_sales_income
▶	34	177000.00	7200.00	169800.00

Query 2 (Analyze sales by category)

SELECT

c.category_name,

SUM(s.quantity) AS quantity_sold,

SUM(s.quantity * s.unit_price - s.discount_amount)

AS net_income,

SUM(s.discount_amount) AS total_discount

FROM Sales s

JOIN Furniture f

ON s.furniture_id = f.furniture_id

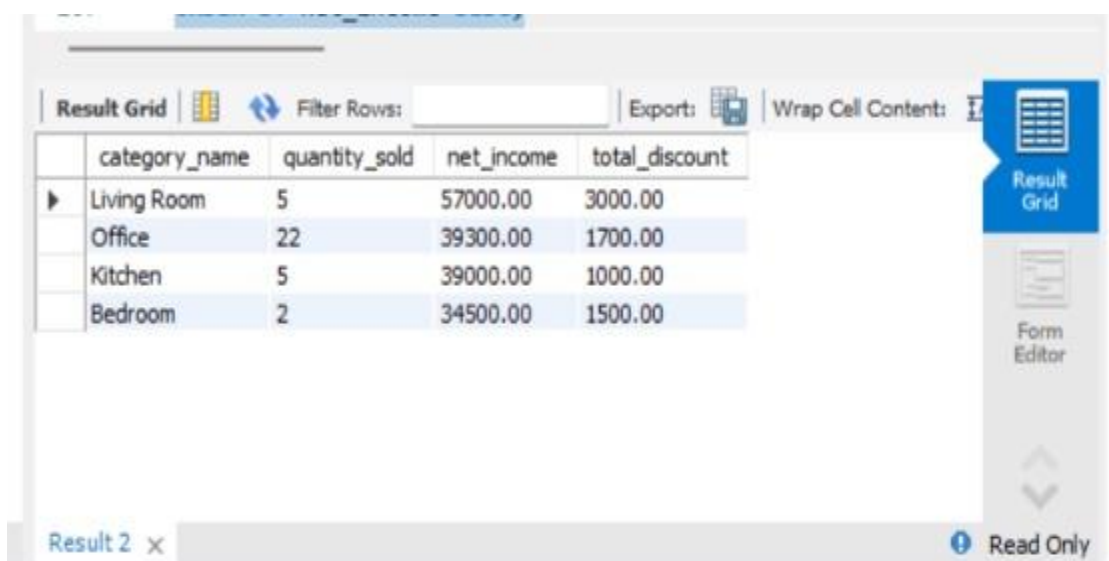
JOIN Categories c

ON f.category_id = c.category_id

GROUP BY c.category_id, c.category_name

ORDER BY net_income DESC;

Result



The screenshot shows a database query result grid with the following data:

	category_name	quantity_sold	net_income	total_discount
▶	Living Room	5	57000.00	3000.00
	Office	22	39300.00	1700.00
	Kitchen	5	39000.00	1000.00
	Bedroom	2	34500.00	1500.00

The interface includes a toolbar with options like 'Filter Rows', 'Export', and 'Wrap Cell Content'. A sidebar on the right contains 'Result Grid' and 'Form Editor' buttons. The bottom status bar shows 'Result 2' and 'Read Only'.

Query 3(Analyze sales by city, region, and state)

SELECT

st.state_name,

r.region_name,

ci.city_name,

SUM(s.quantity) AS quantity_sold,

SUM(s.quantity * s.unit_price - s.discount_amount)

AS net_income,

SUM(s.discount_amount) AS total_discount

FROM Sales s

JOIN Customers cu ON s.customer_id = cu.customer_id

JOIN Cities ci ON cu.city_id = ci.city_id

JOIN Regions r ON ci.region_id = r.region_id

JOIN States st ON r.state_id = st.state_id

GROUP BY

st.state_id, st.state_name,

r.region_id, r.region_name,

ci.city_id, ci.city_name

ORDER BY net_income DESC;

Result



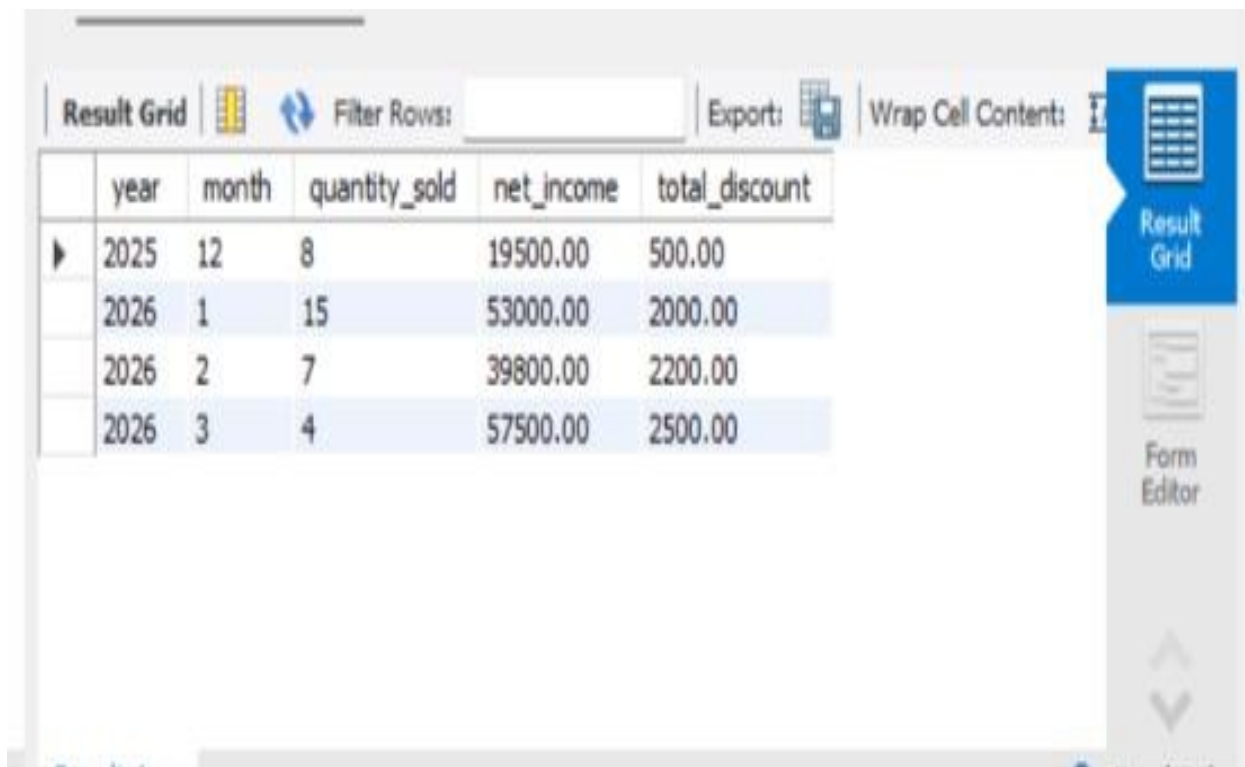
The screenshot shows a database interface with a 'Result Grid' tab selected. The grid displays the results of the SQL query, ordered by net_income in descending order. The columns are state_name, region_name, city_name, quantity_sold, net_income, and total_discount. The first three rows are visible: Madhya Pradesh (Malwa, Indore) with 17 units sold and a net income of 76000.00; Madhya Pradesh (Malwa, Ujjain) with 11 units sold and a net income of 53500.00; and Rajasthan (Jaipur Region, Jaipur) with 6 units sold and a net income of 40300.00. On the right side of the grid, there are buttons for 'Result Grid' and 'Form Editor'.

	state_name	region_name	city_name	quantity_sold	net_income	total_discount
▶	Madhya Pradesh	Malwa	Indore	17	76000.00	3000.00
	Madhya Pradesh	Malwa	Ujjain	11	53500.00	2500.00
	Rajasthan	Jaipur Region	Jaipur	6	40300.00	1700.00

Query 4 (Analyze sales by month and year)

```
SELECT
    td.year,
    td.month,
    SUM(s.quantity) AS quantity_sold,
    SUM(s.quantity * s.unit_price - s.discount_amount)
        AS net_income,
    SUM(s.discount_amount) AS total_discount
FROM Sales s
JOIN TimeDimension td ON s.date_id = td.date_id
GROUP BY td.year, td.month
ORDER BY td.year, td.month;
```

Result



The screenshot shows a software interface with a 'Result Grid' tab selected. The grid displays the results of the SQL query, with columns for year, month, quantity_sold, net_income, and total_discount. The data is sorted by year and month. The interface also includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' option. On the right side, there are buttons for 'Result Grid' and 'Form Editor'.

	year	month	quantity_sold	net_income	total_discount
▶	2025	12	8	19500.00	500.00
	2026	1	15	53000.00	2000.00
	2026	2	7	39800.00	2200.00
	2026	3	4	57500.00	2500.00

Query 5(Find customers whose purchases generated more than ₹20,000 in net income)

SELECT

cu.customer_name,

SUM(s.quantity * s.unit_price - s.discount_amount)

AS net_income

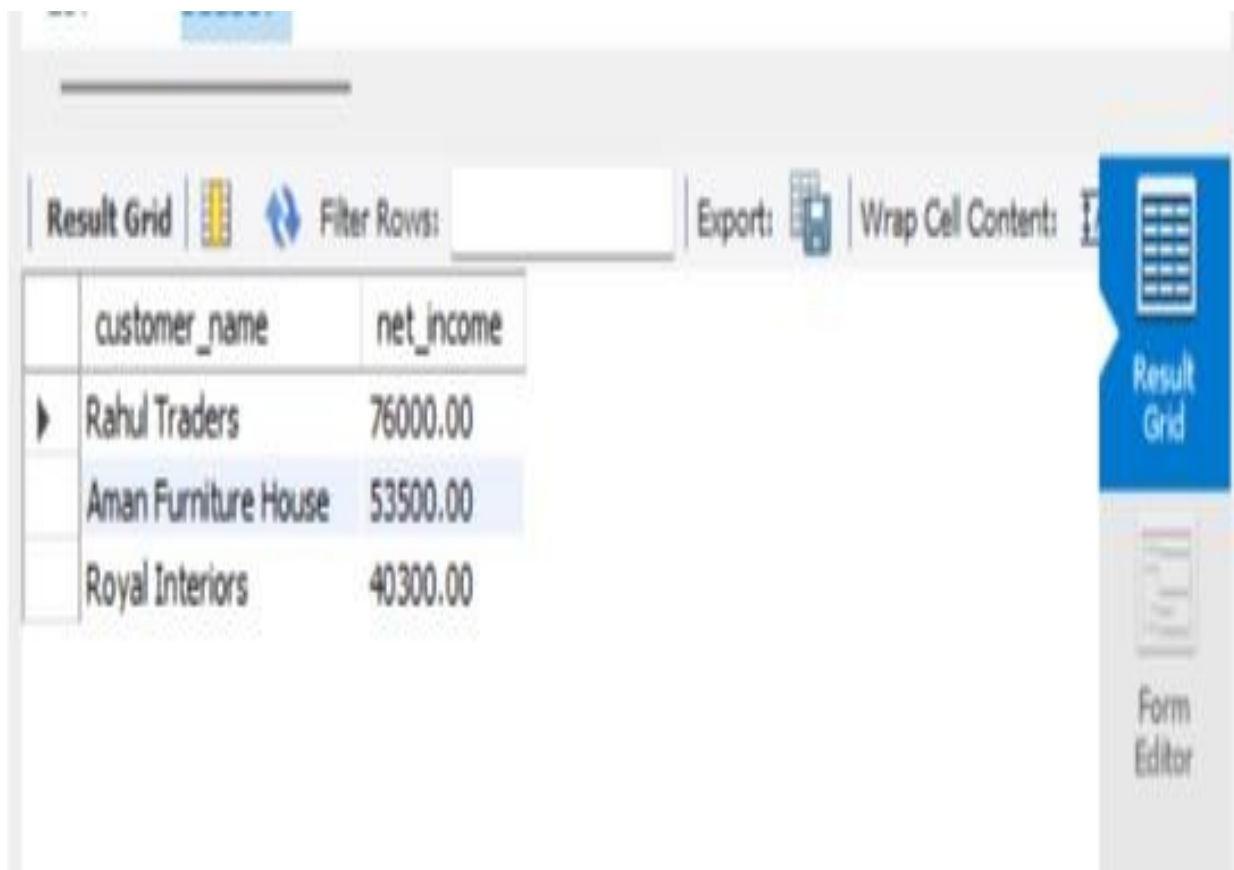
FROM Customers cu

JOIN Sales s ON cu.customer_id = s.customer_id

GROUP BY cu.customer_id, cu.customer_name

HAVING SUM(s.quantity * s.unit_price - s.discount_amount) > 20000;

Result



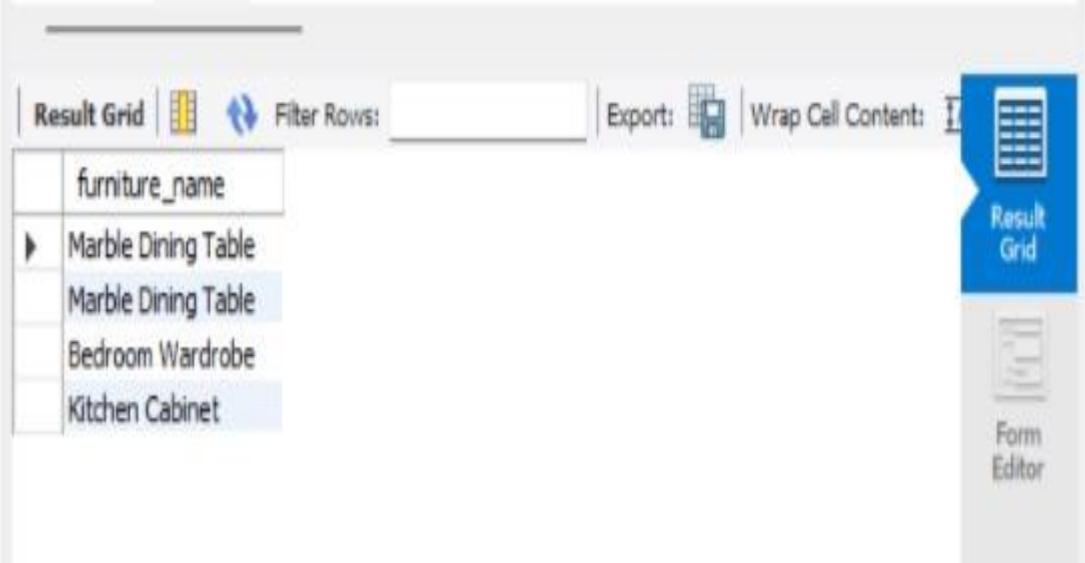
The screenshot shows a database application interface. At the top, there is a toolbar with buttons for 'Result Grid', 'Filter Rows', 'Exports', and 'Wrap Cell Content'. Below the toolbar is a table with two columns: 'customer_name' and 'net_income'. The table contains three rows of data. The first row is 'Rahul Traders' with a net income of 76000.00. The second row is 'Aman Furniture House' with a net income of 53500.00. The third row is 'Royal Interiors' with a net income of 40300.00. On the right side of the interface, there is a vertical sidebar with a 'Result Grid' button and a 'Form Editor' button.

customer_name	net_income
Rahul Traders	76000.00
Aman Furniture House	53500.00
Royal Interiors	40300.00

Query 6(Find furniture priced above the average unit price)

```
SELECT
    f.furniture_name
FROM Furniture f
JOIN Sales s ON f.furniture_id = s.furniture_id
WHERE s.unit_price > (
    SELECT AVG(unit_price)
    FROM Sales
);
```

Result



The screenshot shows a database query result interface. At the top, there is a toolbar with buttons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar is a table with the following data:

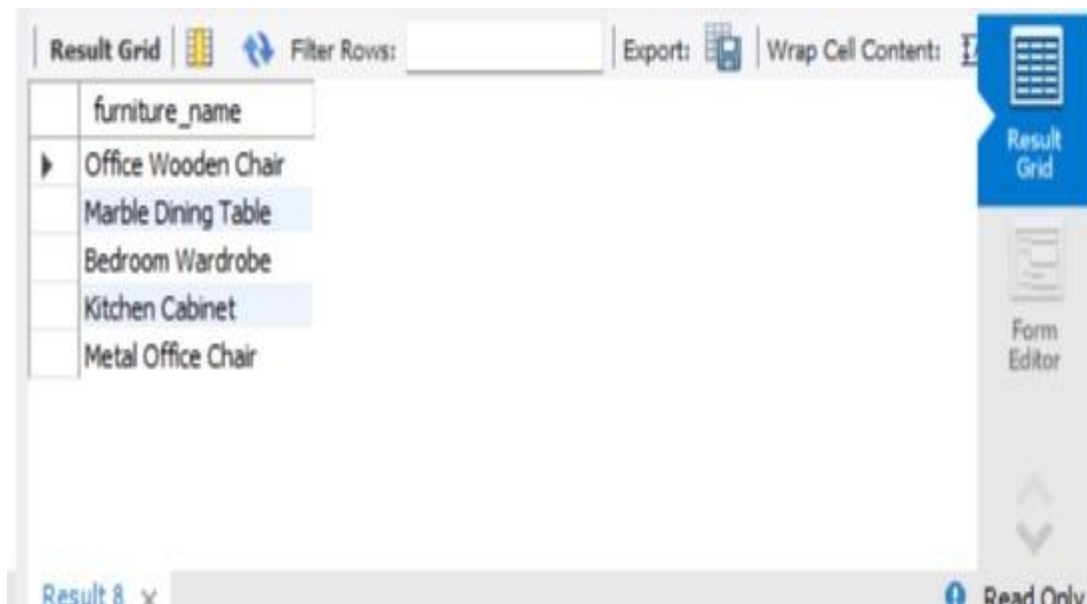
	furniture_name
▶	Marble Dining Table
	Marble Dining Table
	Bedroom Wardrobe
	Kitchen Cabinet

On the right side of the interface, there are two buttons: 'Result Grid' (highlighted in blue) and 'Form Editor'.

Query 7(Find furniture products that were sold)

```
SELECT DISTINCT f.furniture_name  
FROM Furniture f  
JOIN Sales s ON f.furniture_id = s.furniture_id;
```

Result



The screenshot shows a database query result grid. The grid has a single column labeled 'furniture_name'. It contains five rows of data: 'Office Wooden Chair', 'Marble Dining Table', 'Bedroom Wardrobe', 'Kitchen Cabinet', and 'Metal Office Chair'. The first row is highlighted with a blue arrow icon to its left. The interface includes a toolbar at the top with options like 'Filter Rows', 'Export', and 'Wrap Cell Content'. On the right side, there are buttons for 'Result Grid' and 'Form Editor'. The bottom status bar indicates 'Result 8' and 'Read Only'.

furniture_name
Office Wooden Chair
Marble Dining Table
Bedroom Wardrobe
Kitchen Cabinet
Metal Office Chair

Query 8(Demonstrate DML operations)

-- Insert a customer

INSERT INTO Customers

(customer_name, phone, address, city_id)

VALUES ('New Wholesale Customer', '9000000004',
 'Main Market', 1);

-- Update a customer's address

UPDATE Customers

SET address = 'Updated Market Road'

WHERE customer_id = 1;

-- Retrieve customers

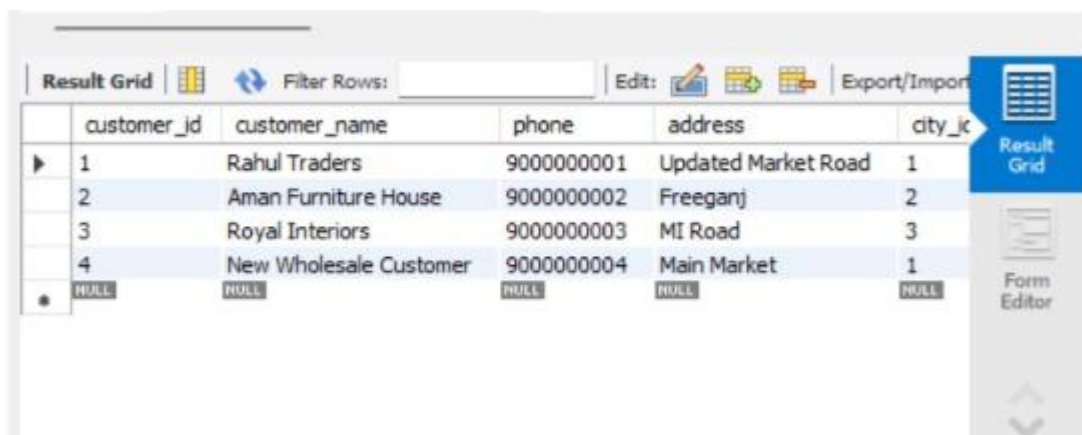
SELECT * FROM Customers;

-- Delete the example customer

DELETE FROM Customers

WHERE customer_name = 'New Wholesale Customer';

Result



	customer_id	customer_name	phone	address	city_id
▶	1	Rahul Traders	9000000001	Updated Market Road	1
	2	Aman Furniture House	9000000002	Freeganj	2
	3	Royal Interiors	9000000003	MI Road	3
	4	New Wholesale Customer	9000000004	Main Market	1
*	NULL	NULL	NULL	NULL	NULL

Query 9(Demonstrate set operators)

```
SELECT customer_name AS name
```

```
FROM Customers
```

```
WHERE city_id = 1
```

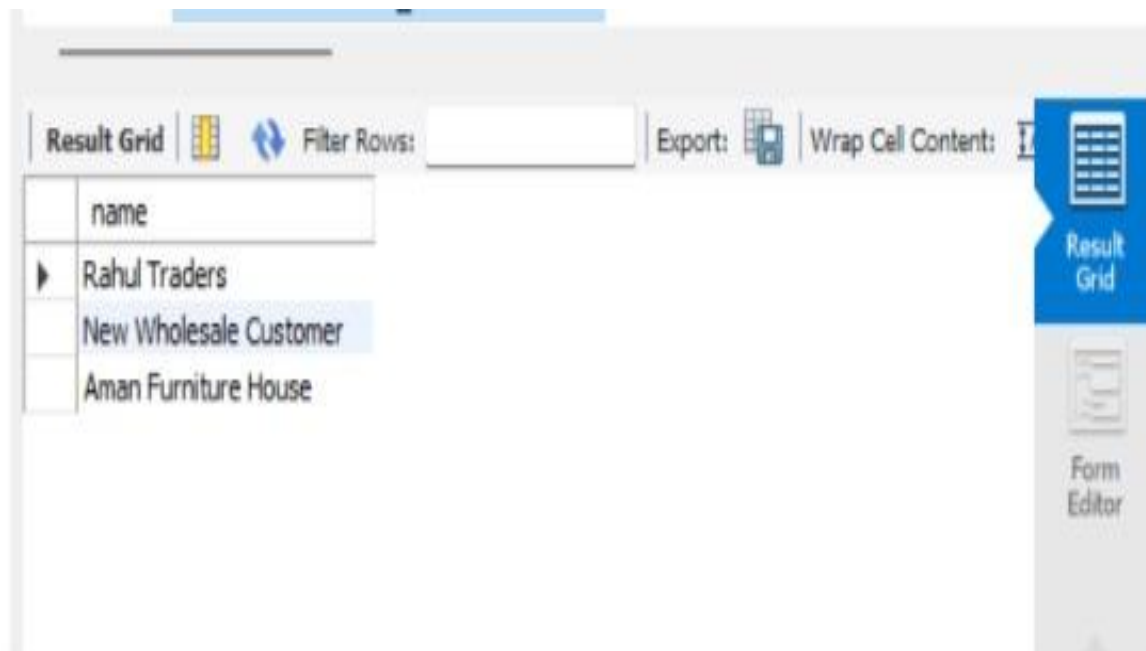
UNION

```
SELECT customer_name AS name
```

```
FROM Customers
```

```
WHERE city_id = 2;
```

Result



The screenshot shows a database application window. At the top, there is a toolbar with buttons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar is a table with the following data:

	name
▶	Rahul Traders
	New Wholesale Customer
	Aman Furniture House

On the right side of the window, there is a vertical sidebar with a 'Result Grid' button and a 'Form Editor' button.

